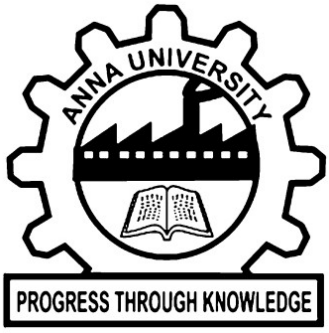
# E-COMMERCE SHOPPING SYSTEM

**A SUMMER INTERNSHIP REPORT**

*Submitted by*

**JENNIS JEROM A - 810023205303**




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# BONAFIDE CERTIFICATE

This is to certify that Summer internship project report **E-COMMERCE SHOPPING SYSTEM** is the bonafide work of **JENNIS JEROM A (810023205303)**, who has satisfactorily completed the course of **IT3711 – SUMMER INTERNSHIP** for **SEVENTH** semester, **B.Tech INFORMATION TECHNOLOGY** during the academic year 2026-2027.

**FACULTY INCHARGE** **HEAD OF THE DEPARTMENT**

# ABSTRACT

This report presents the work carried out during a one-month industrial internship at Fantasy Solution, Tiruchirappalli, from 01 June 2026 to 30 June 2026. The internship combined a structured training programme in Java and enterprise application development with the design and development of an E-Commerce Shopping System.

The training programme covered core and advanced Java, object-oriented programming, collections, multithreading, JDBC, Servlets and JSP, the Spring and Spring Boot frameworks, Hibernate, Spring Security, JUnit testing, Git version control and deployment on Apache Tomcat.

The E-Commerce Shopping System is a web application that lets customers browse a product catalogue, search and filter products, maintain a shopping cart, place orders with a chosen delivery address and payment method, and track their orders. Administrators maintain categories, products, stock and customers, process orders and view sales reports. The system follows a layered architecture with a relational MySQL database, role-based access for administrators and customers, and transactional order processing that keeps stock levels accurate. The report explains the problem, requirements, design with use case and entity-relationship diagrams, implementation, testing strategy and results.

Beyond technical skills, the internship provided exposure to professional practices such as teamwork, documentation, code review and time management, and has strengthened the foundation for a career in software development.

**Keywords:** E-Commerce, Online Shopping, Java, Spring Boot, Hibernate, MySQL, Internship

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CHAPTER 01

# 1. Introduction

## 1.1 Background

An internship is a structured period of professional work experience in which a student applies classroom knowledge to real problems under the guidance of working professionals. Learning happens through doing: the intern joins a team, follows its practices, takes on tasks with real expectations and grows from the outcome. This complements the theoretical foundation built in the classroom.

This report documents the industrial internship completed at Fantasy Solution, Tiruchirappalli, between 01 June 2026 and 30 June 2026, as part of the B.Tech Information Technology programme. The internship combined a focused training programme in Java and enterprise application development with the design and development of a practical product, the E-Commerce Shopping System.

## 1.2 Objectives of the Internship

- To understand how a professional software organisation plans, builds, tests and delivers applications.
- To strengthen core and advanced Java skills, including object-oriented design, collections, multithreading and database connectivity.
- To gain hands-on experience with enterprise technologies such as Servlets, JSP, Spring Boot, Hibernate and Spring Security.
- To apply these skills to a complete project, from requirement analysis and design to implementation, testing and deployment.
- To develop professional habits: communication, documentation, version control, code review and teamwork.

## 1.3 Scope of the Work

The internship covered two connected streams of work. The first was a guided training stream in which fundamental and advanced Java concepts were studied and practised through exercises. The second was the project stream, in which these concepts were applied to build an E-Commerce Shopping System that automates the product catalogue, shopping cart, order placement, payment recording, order tracking, stock control and reporting for an online store. This report concentrates on the knowledge gained, the design decisions taken and the engineering practices followed.

## 1.4 Internship at a Glance

The key details of the internship, as recorded in the official internship annexure, are summarised below.

Table 1.1: Internship details

| Particular | Details |
|---|---|
| Name of the industry | Fantasy Solution, #16, Samnath Plazza, 3rd Floor, Melapudur, Trichy – 620 001 |
| Industrial expert (Joint Supervisor) | Mr. Bragatheesh |
| Qualification of the expert | MCA |
| Designation of the expert | Software Developer |
| Title of the project work | E-Commerce Shopping System |
| Duration of the internship | 01-06-2026 to 30-06-2026 |
| Name of the student | Jennis Jerom A |
| Registration number | 810023205303 |
| Department | B.Tech – Information Technology |

## 1.5 Organisation of the Report

- Chapter 2 presents the profile of Fantasy Solution, its services, mission and vision.
- Chapter 3 describes the technical training programme and the technologies studied.
- Chapters 4 and 5 introduce the E-Commerce Shopping System and analyse its requirements.
- Chapter 6 explains the system design, including architecture, use cases and database design.
- Chapters 7 and 8 cover implementation and testing.
- Chapter 9 walks through the application and reviews the results achieved.
- Chapters 10 and 11 reflect on learning outcomes and challenges, and present the conclusion and future scope.

CHAPTER 02

# 2. Organisation Profile

## 2.1 About Fantasy Solution

Fantasy Solution is an information technology solution and service provider that delivers technology-enabled solutions to businesses, institutions and individuals. The company focuses on software technology and offers industry solutions, product engineering, software products and platforms, and related services.

Its portfolio includes real-time projects, web designing, web hosting, software development and training. Working under the brand commitment of “Beyond Technology”, Fantasy Solution aims to apply information technology to support sustainable development of society while being respected by its employees, shareholders, customers and the community.

## 2.2 Contact Details

Table 2.1: Contact details of the organisation

| Item | Details |
|---|---|
| Organisation | Fantasy Solution |
| Address | No. 16, Samnath Plazza, Third Floor, Near BG Naidu Sweets, Melapudur, Trichy – 620 001 |
| Mobile | 9043095535 |
| Landline | 0431-4971630 |
| Website | www.fantasysolution.in |

## 2.3 Services Offered

Fantasy Solution operates across the full software life cycle: it studies, designs, develops, enhances, customises, implements, maintains and supports different aspects of information technology. Its major service areas are:

- Application development and maintenance, including custom software and enterprise applications.
- E-business solutions, content management systems, payment gateway solutions and time-and-attendance software.
- Professional website design, development and web hosting.
- Mobile application development on the Android platform.
- ERP implementation and consulting, testing and performance engineering.
- IT education and training, including project guidance for students and professionals.

## 2.4 Technology Expertise

Table 2.2: Technology areas handled by the organisation

| Area | Technologies and tools |
|---|---|
| Software development | .NET, PHP and Java |
| Web development | Web designing and development, content management systems, web hosting |
| Mobile | Android application development |
| Data and analytics | Data mining tools, big data development using R and Python |
| Research and simulation | MATLAB for image processing, Network Simulator (NS2) |
| Domain solutions | Electronic health records, payment gateways, appointment reminders, medical transcription |

## 2.5 Mission and Vision

The mission of Fantasy Solution rests on five commitments:

1. Providing high-quality software development services, professional consulting and development outsourcing that improve customers' operations.
2. Making access to information easier and more secure for enterprise business.
3. Improving communication and data exchange between businesses.
4. Providing customers with value for money.
5. Providing employees with meaningful work and opportunities for advancement.

> **NOTE**
> Vision: To be a leading IT company for consulting services and the deployment of best-of-breed business solutions to top-tier domestic and international customers.

## 2.6 Working Environment

The internship began with an orientation in which interns were introduced to the team, the working practices of the organisation and the structure of the programme. Throughout the month the atmosphere was collaborative: senior developers were approachable, doubts were resolved quickly and feedback was given openly. Working inside a professional team helped in understanding workplace etiquette, punctuality, shared responsibility and the importance of clear communication.

CHAPTER 03

# 3. Internship Training Programme

## 3.1 Structure and Approach

The internship ran for thirty days and was organised into progressive phases. The early phase built a strong base in Java, the middle phase introduced data and web technologies, and the later phases applied everything in a project supported by modern frameworks, security, testing and deployment practices. Each concept was introduced through a short explanation followed by coding exercises, so that understanding was always tested by practice.

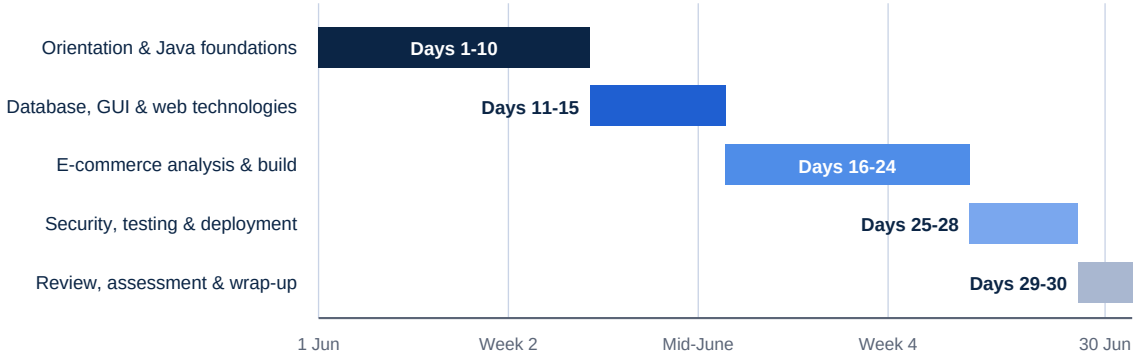



**Figure 3.1:** *Phases of the internship programme (June 2026)*

## 3.2 Core Java and Object-Oriented Programming

The programme started with the Java platform, its features and the architecture of the Java Virtual Machine. Data types, variables, operators and control statements were practised through small programs. The emphasis then moved to object-oriented programming: classes and objects, inheritance, polymorphism, encapsulation and abstraction, followed by interfaces, abstract classes and the role of design patterns in building maintainable code.

Exception handling and debugging formed an important part of this phase. Try-catch blocks, custom exceptions and the debugging tools of the Eclipse IDE were used to locate and fix defects systematically instead of by trial and error.

## 3.3 Advanced Java Concepts

- **Collections Framework:** lists, sets and maps, and choosing the right structure for a problem.
- **Generics:** type-safe, reusable classes and methods.
- **File I/O:** reading from and writing to different kinds of files.
- **Multithreading and concurrency:** thread creation, synchronisation and inter-thread communication.
- **Networking:** socket programming and a simple client-server chat application.

## 3.4 Database and Web Technologies

Database connectivity was studied with JDBC: connecting Java applications to relational databases, executing SQL queries, handling result sets and implementing CRUD (Create, Read, Update, Delete) operations in a prototype application. JavaFX was used to build simple graphical interfaces. Web development began with Servlets and JavaServer Pages (JSP), followed by an overview of Java EE capabilities such as web services and the Java Persistence API (JPA).

## 3.5 Frameworks: Spring and Hibernate

The framework phase introduced the Spring ecosystem. Dependency injection and aspect-oriented programming were practised by configuring a Spring project and its beans. Spring Boot was then used to produce standalone, production-quality applications and a RESTful web service. Hibernate was studied for object-relational mapping and combined with Spring Boot to build a full application with a Hibernate-backed persistence layer. Spring Security introduced authentication and authorisation for protecting application features.

## 3.6 Quality, Version Control and Deployment

Engineering practices received equal attention. JUnit was used for unit and integration testing, Git for version control and teamwork, and Java web applications were deployed on Apache Tomcat. The programme closed with documentation, project presentations before mentors and peers, and a review of the experience.

## 3.7 Technology Summary

Table 3.1: Technologies studied and their use in the project

| Technology | What was learnt | Role in the E-Commerce Shopping System |
|---|---|---|
| Core Java, OOP | Classes, inheritance, interfaces, exceptions | Domain model: User, Category, Product, CartItem, Order, Payment |
| Collections, Generics | Lists, sets, maps, type-safe code | In-memory handling of cart contents, search results and reports |
| JDBC, SQL | Queries, result sets, CRUD | Foundation for the database layer |
| Servlets, JSP | Request handling, dynamic pages | Storefront pages and form handling |
| Spring, Spring Boot | Dependency injection, REST services | Controllers, services and application configuration |
| Hibernate, JPA | Object-relational mapping | Mapping entities to MySQL tables |
| Spring Security | Authentication, authorisation | Role-based access for administrator and customer; password hashing |
| JUnit | Unit and integration testing | Verification of pricing, stock and order rules |
| Git, Tomcat | Version control, deployment | Source management and application hosting |

CHAPTER 04

# 4. Project Overview

## 4.1 Introduction to the E-Commerce Shopping System

Online shopping has changed the way people buy everyday goods. A customer expects to browse products from anywhere, compare prices, collect items in a cart, pay securely and follow the delivery of the order. A seller, in turn, needs to keep the catalogue and stock accurate, process orders quickly and understand sales trends. An E-Commerce Shopping System is a software application that supports both sides of this activity.

The project assigned during the internship was to design and develop such a system using the Java technology stack studied in the training programme. It provided an opportunity to follow a complete software development life cycle on a problem that is easy to understand yet rich enough to require careful design.

## 4.2 Problem Statement

Small and medium sellers who depend on phone calls, messaging applications and paper registers struggle with an out-of-date product list, orders that are missed or entered twice, stock that does not match what is on the shelf, manual calculation of price, tax and delivery charges, and no clear record of payments and deliveries. The problem addressed by this project is to provide a secure, easy-to-use and reliable web-based store that replaces these manual steps with an integrated digital workflow.

## 4.3 Existing System and its Limitations

- **Limited reach:** products can be seen only by customers who visit the shop or receive a message.
- **Manual orders:** orders taken over phone or chat are written down by hand and are easy to lose or duplicate.
- **Stock mismatch:** stock is updated by hand, so items are sold that are no longer available.
- **Error-prone billing:** totals, discounts, tax and delivery charges are calculated manually.
- **Poor order tracking:** customers cannot see the status of an order without calling the seller.
- **Limited reporting:** preparing sales and stock statistics takes hours.

## 4.4 Proposed System

The proposed E-Commerce Shopping System stores all information in a central relational database and exposes it through a role-based web interface. Customers register, browse and search the catalogue, fill a cart and place orders; administrators maintain the catalogue, stock and orders. Stock is checked and reduced automatically when an order is placed, prices, discounts, tax and delivery charges are calculated by the system, and reports are generated on demand.

## 4.5 Objectives of the Project

- Maintain a complete, searchable catalogue of categories and products with price and stock.
- Allow customers to register, log in, browse, search, filter and sort products.
- Provide a shopping cart and a simple checkout with delivery address and payment method.
- Calculate subtotal, discount, GST, delivery charge and final amount automatically.
- Keep stock accurate by validating and reducing it inside a single transaction when an order is placed.
- Let customers track orders and let administrators update order status.
- Provide secure, role-based access and generate sales, stock and order reports.

## 4.6 Scope and Limitations

The system covers catalogue management, customer accounts, cart, checkout, order processing, payment recording, order tracking and reporting for a single online store. Items outside the present scope, such as integration with a live payment gateway, multi-vendor marketplaces, product recommendations and courier integration, are listed as future enhancements in Chapter 11.

## 4.7 Feasibility Study

Table 4.1: Feasibility analysis

| Type | Assessment |
|---|---|
| Technical | The project uses mature, widely supported open technologies (Java, Spring Boot, Hibernate, MySQL) for which skills were acquired during the training programme. |
| Economic | All software components are open source and the application runs on modest hardware, so development and running costs are low compared with the manual effort saved. |
| Operational | The interface follows familiar shopping patterns (catalogue, cart, checkout), so customers and staff need very little training to use it. |
| Schedule | The scope was planned to fit the four-week internship by building the system module by module. |

CHAPTER 05

# 5. System Analysis and Requirements

## 5.1 Development Methodology

An iterative approach was followed. The requirements were first understood in discussion with the industry guide, then the system was designed at a high level and implemented in small modules. Each module was tested and reviewed before the next one was started, which kept the risk of large rework low and allowed feedback from the mentor to be absorbed early.

## 5.2 Functional Requirements

Table 5.1: Functional requirements

| ID | Requirement | Role |
|---|---|---|
| FR-01 | Users shall register, log in with email and password, and log out securely. | All |
| FR-02 | The system shall add, update, delete and view categories and products with price and stock. | Admin |
| FR-03 | Customers shall browse, search, filter and sort products by category and price. | Customer |
| FR-04 | Customers shall add products to a cart, change quantities and remove items. | Customer |
| FR-05 | Customers shall manage delivery addresses and place an order from the cart through checkout. | Customer |
| FR-06 | The system shall validate stock and compute subtotal, discount, GST, delivery charge and total. | System |
| FR-07 | The system shall record the payment method (UPI, card, cash on delivery) and confirm the order. | Customer, System |
| FR-08 | Customers shall view order history, track order status and cancel an order before it is shipped. | Customer |
| FR-09 | Administrators shall view orders, update order status and manage customers. | Admin |
| FR-10 | The system shall generate sales, stock and order reports. | Admin |

## 5.3 Non-Functional Requirements

Table 5.2: Non-functional requirements

| Attribute | Requirement |
|---|---|
| Security | Passwords are stored as salted hashes; every page and action is protected by role-based authorisation. |
| Usability | Clean, consistent screens with clear validation messages and minimal clicks to buy a product. |
| Reliability | Order placement and stock updates are transactional, so records never become inconsistent. |
| Performance | Catalogue searches and listings respond quickly through indexed database columns and paging. |
| Maintainability | Layered architecture with separate controller, service and repository code. |
| Portability | Runs on any platform that supports Java and MySQL, with deployment on Apache Tomcat. |

## 5.4 Business Rules

The business rules are kept as configuration values so that the store owner can change them without modifying code. The values below were used as the sample configuration during development.

Table 5.3: Sample business rules

| Rule | Sample value |
|---|---|
| Order status flow | PLACED, PACKED, SHIPPED, DELIVERED. An order can be CANCELLED only before it is shipped. |
| GST | 18% of the amount after discount |
| Delivery charge | Free when the amount after discount is Rs. 500 or more; otherwise Rs. 40 |
| Discount coupon | SAVE10 gives 10% off the subtotal; one coupon per order |
| Cart quantity | At least 1 and at most 10 units of a product per order |
| Stock | Quantity ordered cannot exceed available stock; stock is reduced when the order is placed and restored on cancellation |
| Customer email | Must be unique for every account |

## 5.5 Hardware and Software Requirements

Table 5.4: Development and deployment environment

| Component | Specification |
|---|---|
| Processor | Intel Core i3 or higher (or equivalent) |
| Memory | 4 GB RAM minimum, 8 GB recommended |
| Storage | 10 GB of free disk space |
| Operating system | Windows 10/11 or Linux |
| Language and runtime | Java Development Kit (JDK) 17 or later |
| Frameworks | Spring Boot, Spring MVC, Spring Security, Hibernate / JPA |
| Database | MySQL 8 |
| Web server | Apache Tomcat |
| IDE and tools | Eclipse IDE, Git, Maven, JUnit |
| Browser | Any modern browser such as Chrome, Edge or Firefox |

CHAPTER 06

# 6. System Design

## 6.1 Design Approach

The design separates the application into layers so that each part has a single responsibility. The user interface only presents data, controllers receive and validate requests, the service layer enforces business rules, and the data access layer talks to the database. This separation makes the system easier to understand, test and extend, and it directly reflects the layered architecture studied during the Spring and Hibernate training.

## 6.2 System Architecture

The E-Commerce Shopping System follows a layered, three-tier architecture. A security layer sits beside the controllers and checks the identity and role of every request before it reaches the business logic.

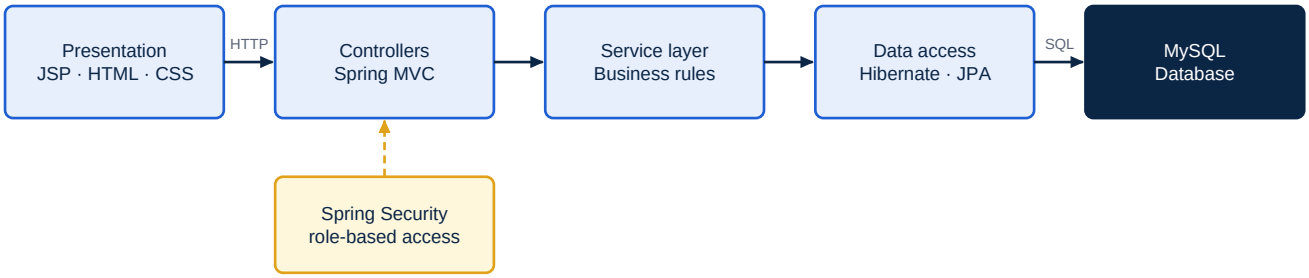



**Figure 6.1:** *Layered architecture of the E-Commerce Shopping System*

## 6.3 Use Case Diagram

The use case diagram shows how the two types of users interact with the system. All users must log in; beyond that, each role has its own set of permitted functions.

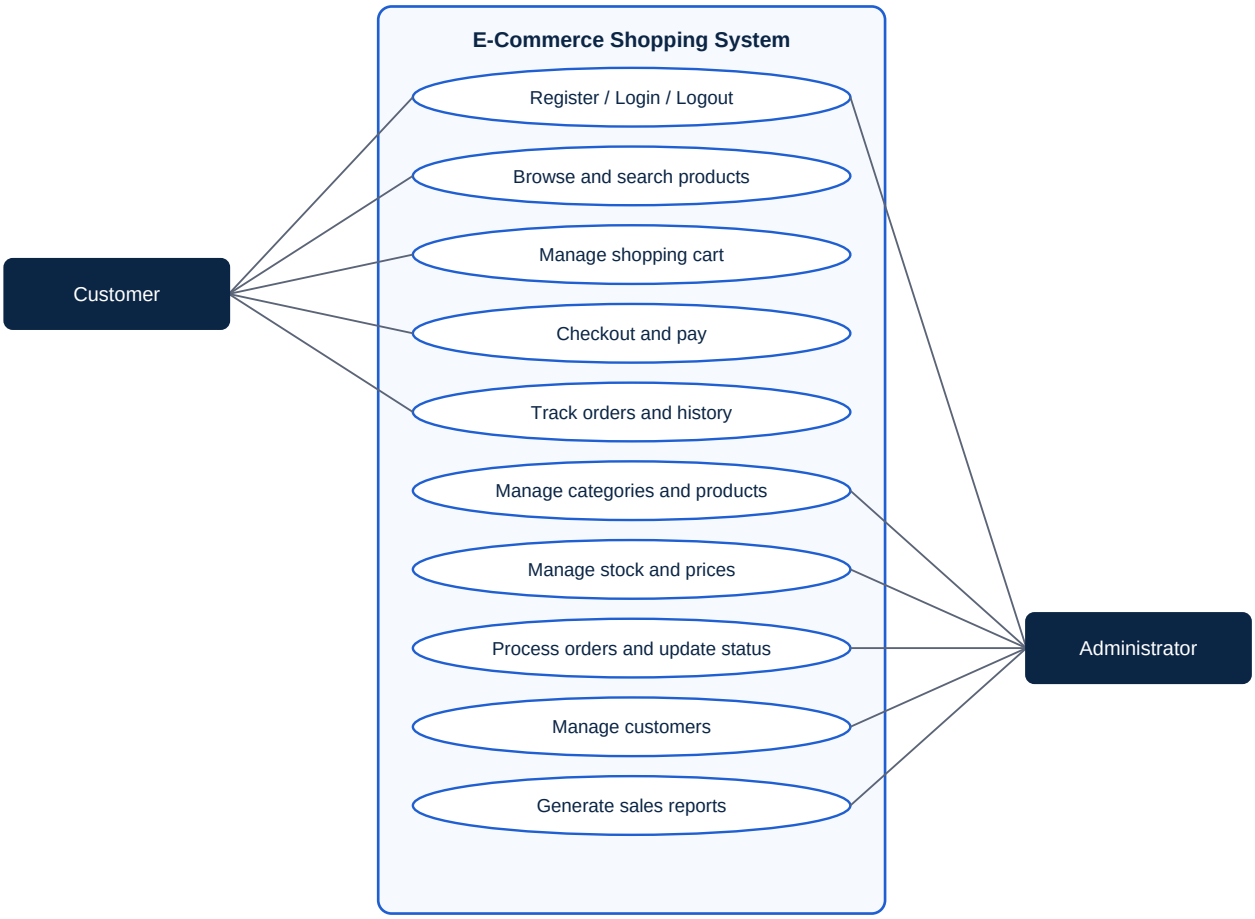



**Figure 6.2:** *Use case diagram*

## 6.4 Entity-Relationship Diagram

The database is designed around eight entities. A category groups many products; a customer (user) owns addresses, a cart and orders; an order contains many order items, each pointing to a product, and has one payment record.

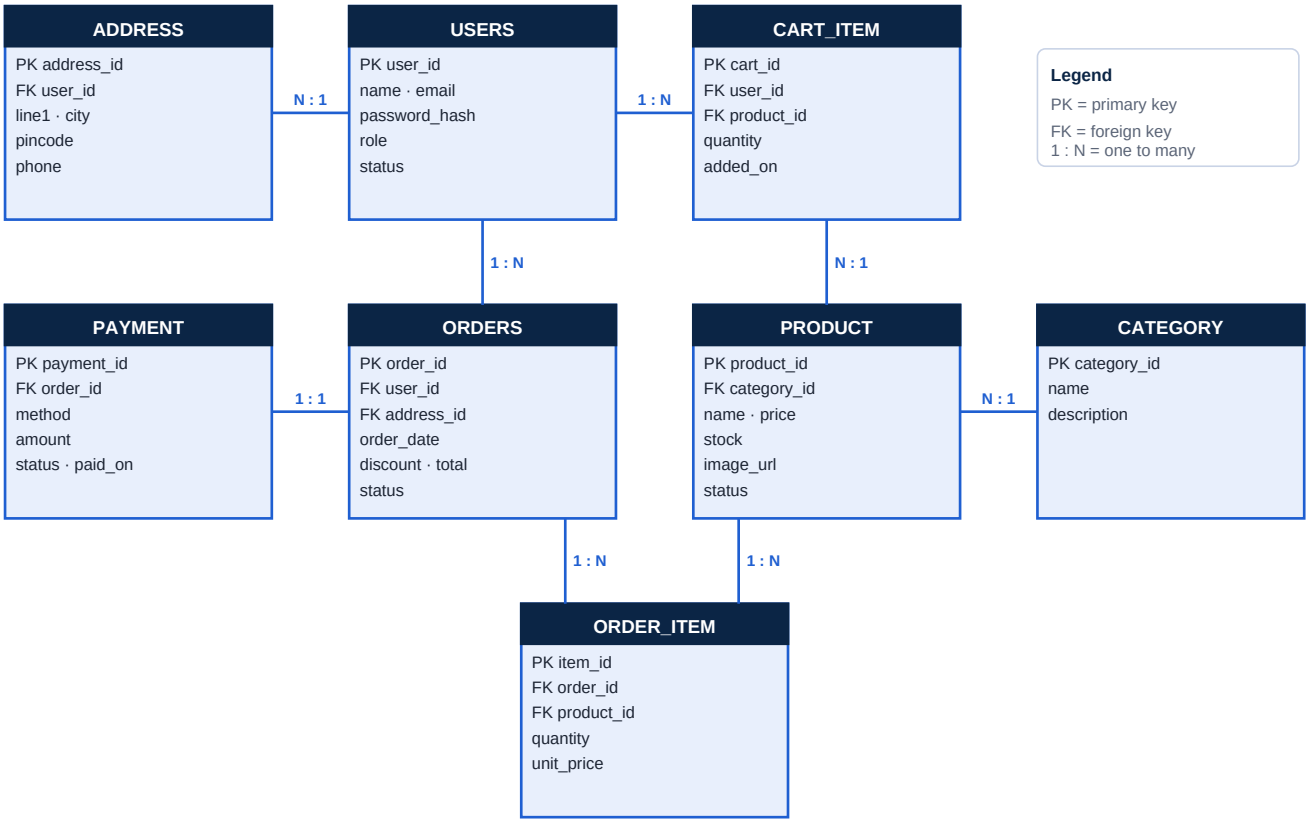



**Figure 6.3:** *Entity-relationship diagram*

## 6.5 Database Design

The two central tables are described below; the remaining tables follow the attributes shown in the entity-relationship diagram. Primary keys are indexed automatically, and foreign keys enforce referential integrity so that an order item can never refer to a missing order or product.

Table 6.1: Table PRODUCT

| Column | Type | Constraint / description |
|---|---|---|
| product_id | INT | Primary key, auto-increment |
| category_id | INT | Foreign key to CATEGORY |
| name | VARCHAR(150) | Not null, indexed for search |
| description | TEXT | Product details shown on the product page |
| price | DECIMAL(10,2) | Unit price, greater than 0 |
| stock | INT | Available quantity, never negative |
| image_url · status | VARCHAR | Image path; status ACTIVE or INACTIVE |

Table 6.2: Table ORDERS

| Column | Type | Constraint / description |
|---|---|---|
| order_id | INT | Primary key, auto-increment |
| user_id | INT | Foreign key to USERS |
| address_id | INT | Foreign key to ADDRESS (delivery address) |
| order_date | DATETIME | Date and time at which the order was placed |
| discount | DECIMAL(10,2) | Coupon discount applied to the order |
| total | DECIMAL(10,2) | Final amount payable including GST and delivery |
| status | VARCHAR(15) | PLACED, PACKED, SHIPPED, DELIVERED or CANCELLED |

## 6.6 Process Flow: Checkout, Payment and Order Placement

Order placement is the heart of the system. Stock is verified for every item in the cart; the system then applies the coupon and computes discount, GST, delivery charge and total, and finally confirms the order only when the payment step succeeds. On success the order is created, stock is reduced and the cart is cleared.

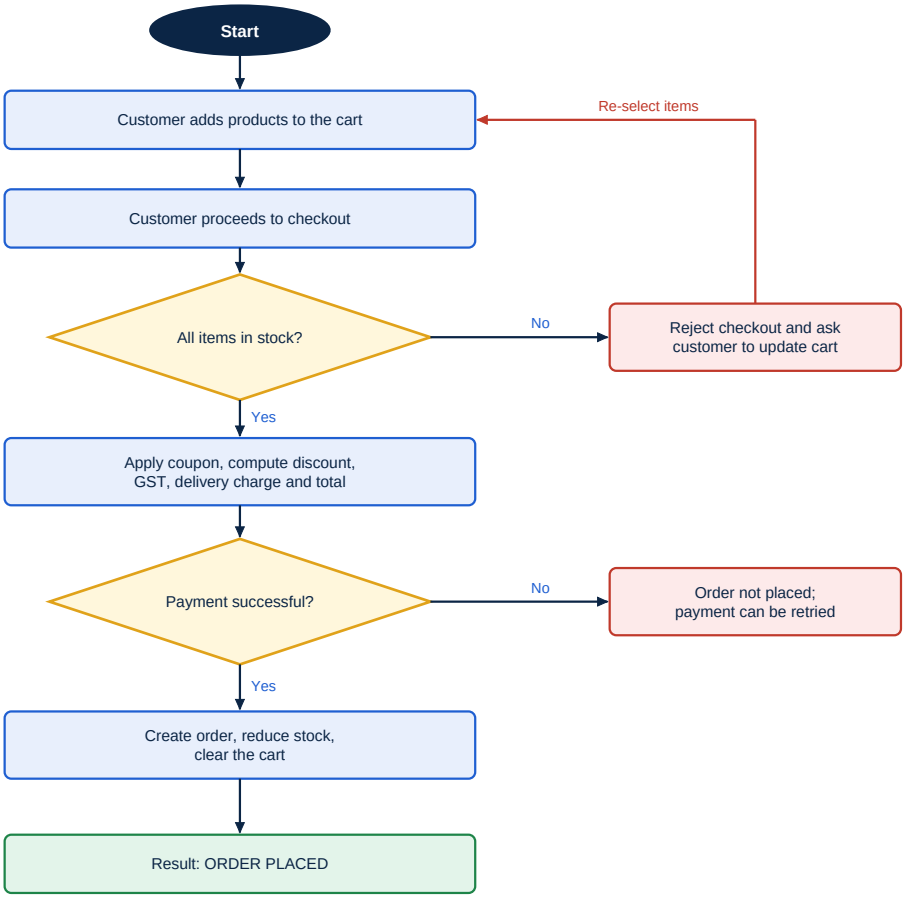



**Figure 6.4:** *Flowchart of checkout, payment and order placement*

CHAPTER 07

# 7. Implementation

## 7.1 Development Environment

The application was developed in Java using the Eclipse IDE. Spring Boot provided the application skeleton, embedded configuration and REST support; Hibernate with JPA handled persistence to a MySQL database; the web pages were built with HTML, CSS, JavaScript and JSP; Maven managed dependencies; and Git tracked every change to the source code. The application was packaged and deployed on Apache Tomcat.

## 7.2 Project Structure

The source code is organised by responsibility so that a developer can quickly find where a change belongs.

```
ecommerce-shopping/
    controller/   request handling (Auth, Product, Cart, Order, Report)
    service/      business rules (pricing, stock, order processing)
    repository/   Spring Data / Hibernate data access
    model/        JPA entities (User, Product, CartItem, Order, Payment)
    security/     authentication and role configuration
    templates/    JSP / HTML views
    resources/    application properties, SQL scripts
```

## 7.3 Module Description

Table 7.1: Modules of the E-Commerce Shopping System

| Module | Description |
|---|---|
| Authentication and access | Registration, login and logout, password hashing and role-based authorisation for administrator and customer. |
| Category and product management | Add, edit, delete and deactivate categories and products; maintain price, stock and images. |
| Catalogue and search | Product listing with keyword search, category filter, price sort and paging. |
| Shopping cart | Add, update and remove cart items with stock and quantity checks; live subtotal. |
| Checkout and orders | Address selection, coupon, price calculation and transactional order placement with stock reduction. |
| Payment | Records payment method and status for each order; supports UPI, card and cash on delivery. |
| Order tracking and management | Customers view history and status and cancel before shipping; administrators update status. |
| Reports | Sales, stock and order reports with export option. |

## 7.4 Key Implementation Logic

Two pieces of logic illustrate how business rules were implemented. The price calculation is kept in the service layer so that it can be unit tested independently of the web layer, and uses BigDecimal to avoid rounding errors in money values.

```
public BigDecimal calculateTotal(BigDecimal subtotal, BigDecimal discount) {
    BigDecimal taxable = subtotal.subtract(discount);
    BigDecimal gst = taxable.multiply(new BigDecimal("0.18"));
    BigDecimal delivery = taxable.compareTo(new BigDecimal("500")) >= 0
            ? BigDecimal.ZERO : new BigDecimal("40");
    return taxable.add(gst).add(delivery).setScale(2, RoundingMode.HALF_UP);
}
```

Placing an order runs inside a single database transaction, so either the order is created, the stock reduced and the cart cleared together, or none of these happens.

```
@Transactional
public Order placeOrder(Long userId, Long addressId) {
    List<CartItem> cart = cartRepo.findByUserId(userId);
    if (cart.isEmpty()) throw new EmptyCartException();
    Order order = new Order(userId, addressId, LocalDateTime.now());
    for (CartItem ci : cart) {
        Product p = productRepo.findById(ci.getProductId()).orElseThrow();
        if (p.getStock() < ci.getQuantity())
            throw new OutOfStockException(p.getName());
        p.setStock(p.getStock() - ci.getQuantity());
        order.addItem(new OrderItem(p, ci.getQuantity(), p.getPrice()));
    }
    order.setStatus(OrderStatus.PLACED);
    cartRepo.deleteAll(cart);
    return orderRepo.save(order);
}
```

> **NOTE**
> The snippets above are simplified illustrations of the approach used in the project. Class and method names in the final source code may differ slightly.

## 7.5 Security Implementation

- **Authentication:** users sign in with credentials verified against salted password hashes.
- **Authorisation:** Spring Security restricts every URL and action by role, so customers cannot open administrator pages.
- **Input validation:** server-side validation prevents invalid quantities, negative prices, duplicate emails and malformed addresses.
- **SQL injection protection:** all database access uses parameterised queries through JPA and Hibernate.
- **Session handling:** sessions are invalidated on logout and expire after a period of inactivity.

## 7.6 Version Control and Deployment

Source code was managed with Git, with regular commits describing each change so that the history of the project is easy to follow and any change can be reverted if required. For deployment, the application was packaged as a web archive and deployed on Apache Tomcat, with the MySQL database created from the SQL scripts included in the project.

CHAPTER 08

# 8. Testing

## 8.1 Objectives of Testing

Testing was carried out to confirm that the system behaves as specified, that business rules are enforced under normal and unusual inputs, and that unauthorised users cannot reach restricted functions. Test cases were designed from the functional requirements in Chapter 5 so that every requirement is covered by at least one scenario.

## 8.2 Testing Strategy

Table 8.1: Levels of testing

| Level | Focus | Tools / method |
|---|---|---|
| Unit testing | Individual methods such as price calculation and stock validation | JUnit |
| Integration testing | Interaction between controller, service and database layers | JUnit, Spring Boot test support |
| System testing | Complete workflows such as browsing, cart, checkout and order tracking | Manual test scripts |
| Security testing | Role restrictions, session expiry and invalid input handling | Manual verification |
| User acceptance testing | Usability and correctness from the point of view of the mentor and users | Demonstration and feedback |

## 8.3 Sample Test Cases

Table 8.2: Representative test cases and expected results

| ID | Scenario | Input / condition | Expected result |
|---|---|---|---|
| TC-01 | Valid login | Correct email and password | User reaches the home page or dashboard for the role |
| TC-02 | Invalid login | Wrong password | Error message; access denied |
| TC-03 | Add product (admin) | All mandatory fields filled | Product saved and visible in the catalogue |
| TC-04 | Search product | Keyword “shirt” | Only matching products are listed |
| TC-05 | Add to cart | In-stock product, quantity 2 | Cart updated with correct line total |
| TC-06 | Quantity above stock | Quantity greater than available stock | Request rejected with a clear message |
| TC-07 | Place order | Valid cart and delivery address | Order created; stock reduced; cart cleared |
| TC-08 | Empty cart checkout | Checkout with no items | Checkout blocked with a message |
| TC-09 | Total computation | Subtotal Rs. 1,000; coupon SAVE10; GST 18% | Total Rs. 1,062.00 (delivery free) |
| TC-10 | Role restriction | Customer opens an admin URL | Access denied |

## 8.4 Defect Handling

Defects found during testing were recorded with the steps needed to reproduce them, fixed in the relevant layer, and re-tested together with the surrounding functionality to make sure that the fix had not affected anything else. Typical defects of this kind are rounding errors in tax amounts, missing validation messages and menu options displayed to the wrong role. Using JUnit for the business rules allowed such defects to be caught quickly and kept from returning.

CHAPTER 09

# 9. Application Walkthrough and Results

This chapter walks through the main screens of the E-Commerce Shopping System. Customers see a storefront with a top navigation bar; administrators see a left navigation menu that shows only the options permitted for their role. The figures show the sample data used for demonstration.

## 9.1 Login Page

Every user begins at the login page, which asks for an email and password verified against salted password hashes. New customers can register from the same page. After successful authentication the user is taken to a page that matches the role, and unauthorised users are denied access to every other page.

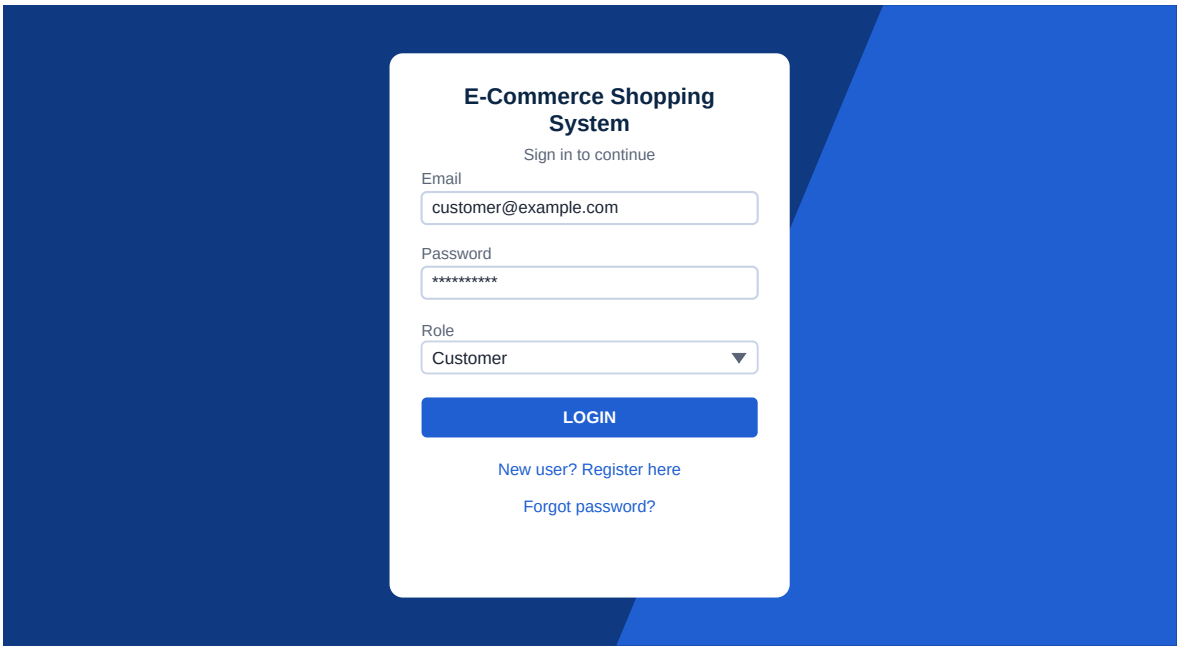



**Figure 9.1:** *Login page*

## 9.2 Product Catalogue

The catalogue lists products with image, name, price and rating. Customers can search by keyword, filter by category, sort by price and move through pages; the Add to Cart button places a product in the cart in one click.

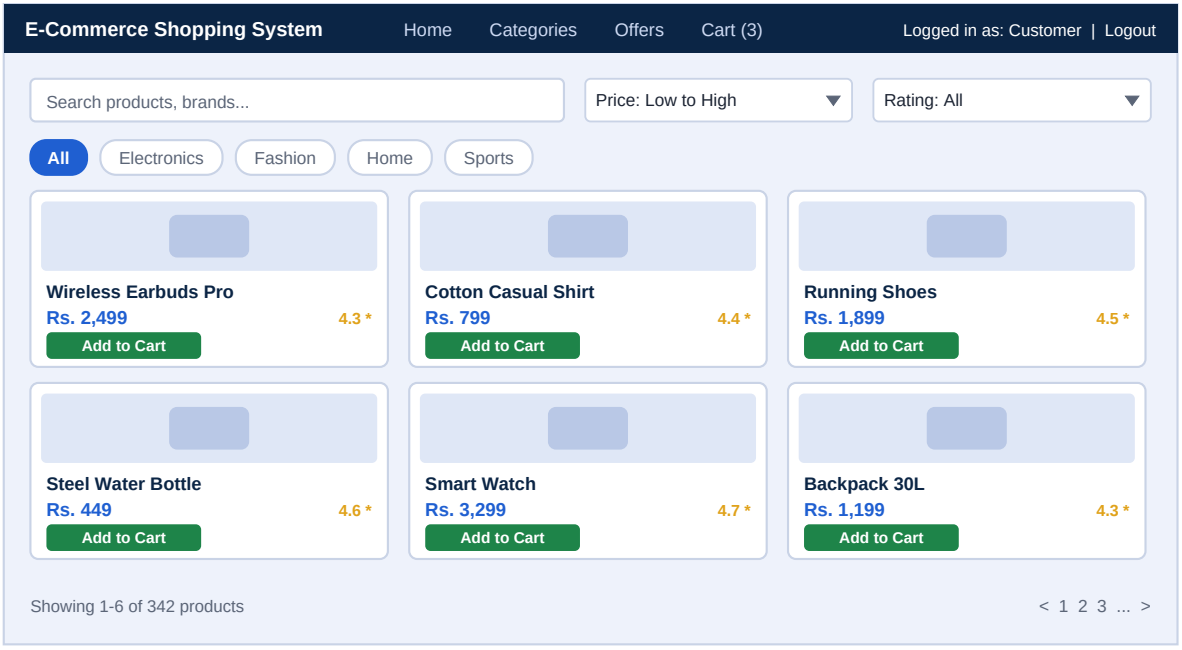



**Figure 9.2:** *Product catalogue with search, filters and Add to Cart*

## 9.3 Shopping Cart

The cart lists the selected products with quantity controls and line totals. Quantities above the available stock are rejected. The order summary shows the subtotal, the coupon discount, GST, the delivery charge and the final amount, and updates as soon as an item is changed.

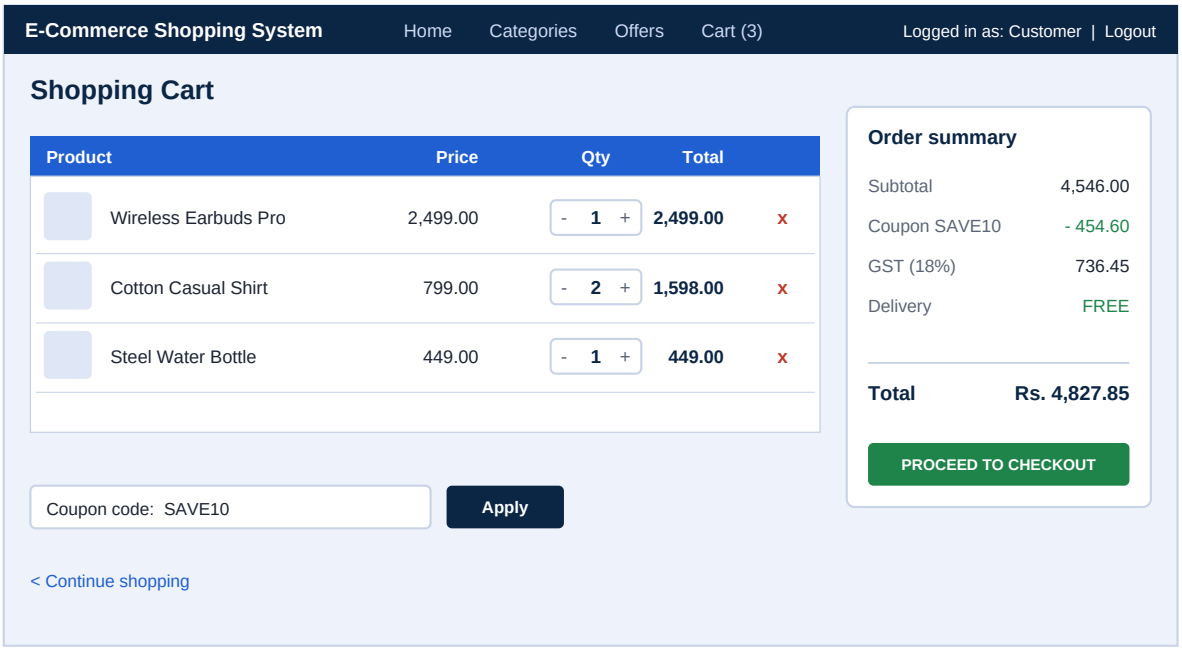



**Figure 9.3:** *Shopping cart with order summary*

## 9.4 Checkout and Payment

At checkout the customer selects a delivery address and a payment method. The system re-validates stock for every item and then places the order inside one transaction, so stock and order records always agree.

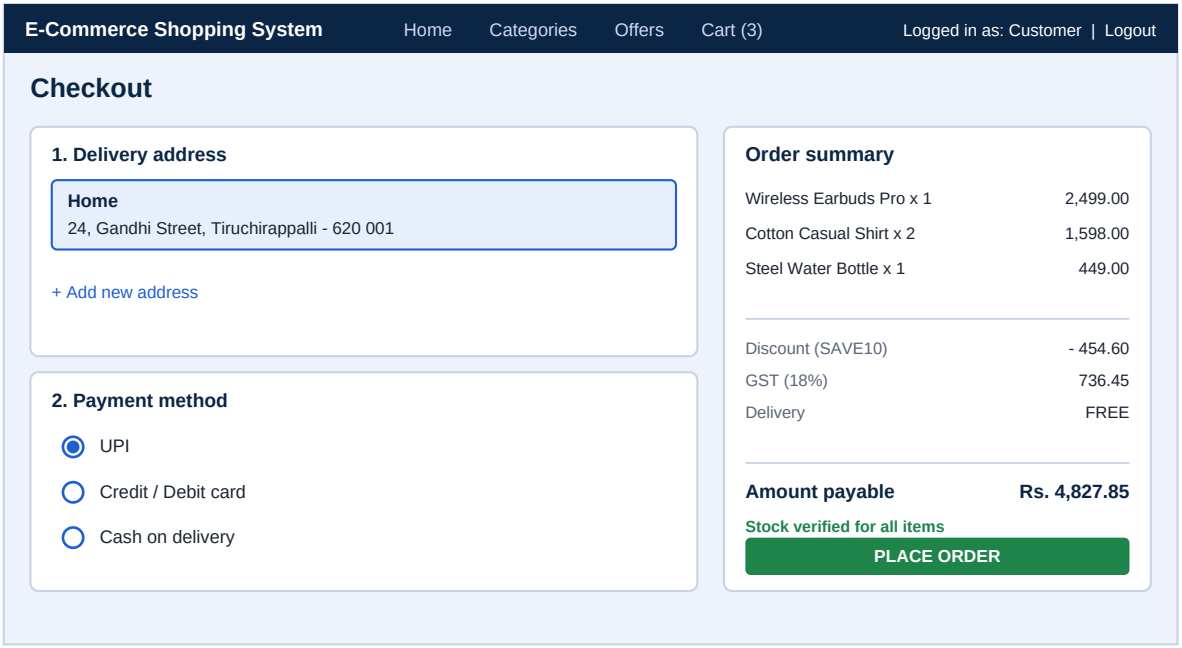



**Figure 9.4:** *Checkout with delivery address and payment method*

## 9.5 Order History and Tracking

Customers see all their orders with a progress tracker (Placed, Packed, Shipped, Delivered). An order that is still in the Placed state can be cancelled, which restores the stock.

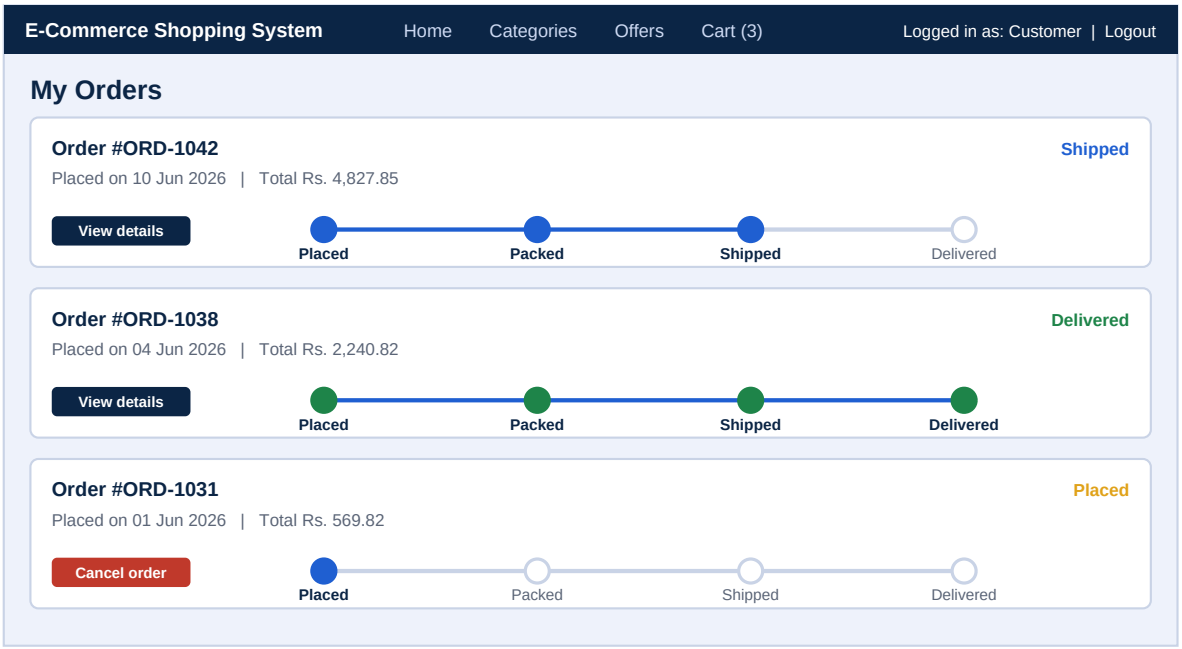



**Figure 9.5:** *Order history and order tracking*

## 9.6 Administrator Dashboard

The dashboard gives the administrator a quick picture of the store: summary tiles for products, customers, orders and revenue, shortcuts to common tasks and a recent activity list that includes low-stock alerts.

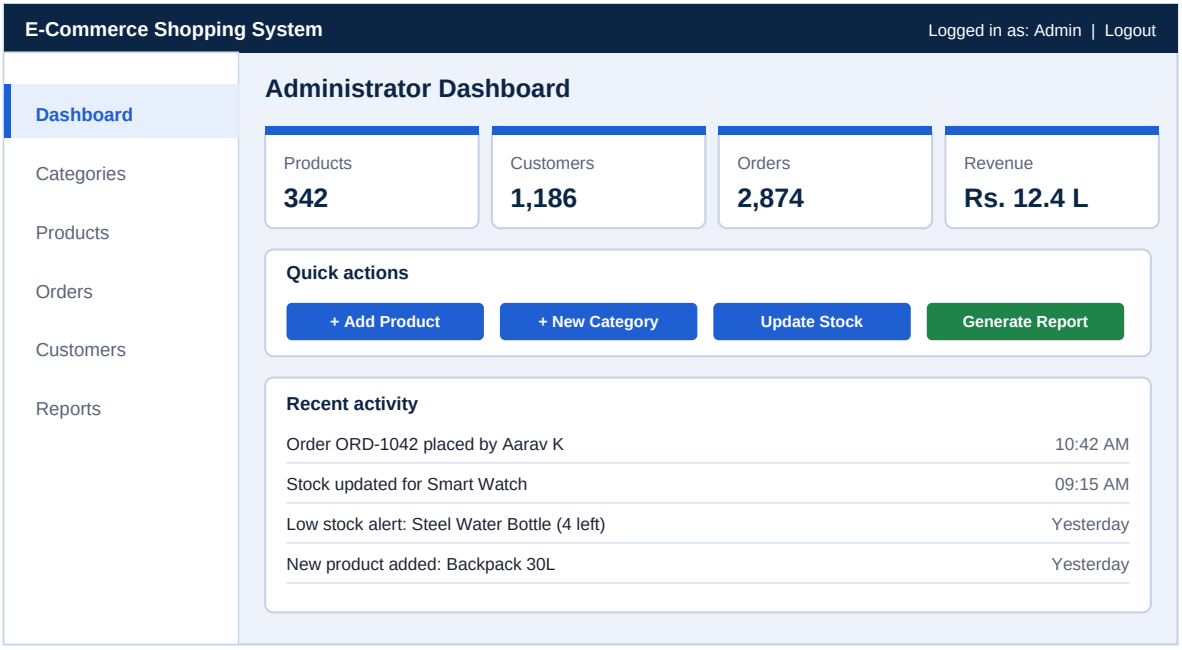



**Figure 9.6:** *Administrator dashboard with summary tiles and quick actions*

## 9.7 Order Management and Reports

Administrators review incoming orders and move each one through its status. The reports module summarises sales, stock and orders for a selected period and can be exported as PDF or Excel.

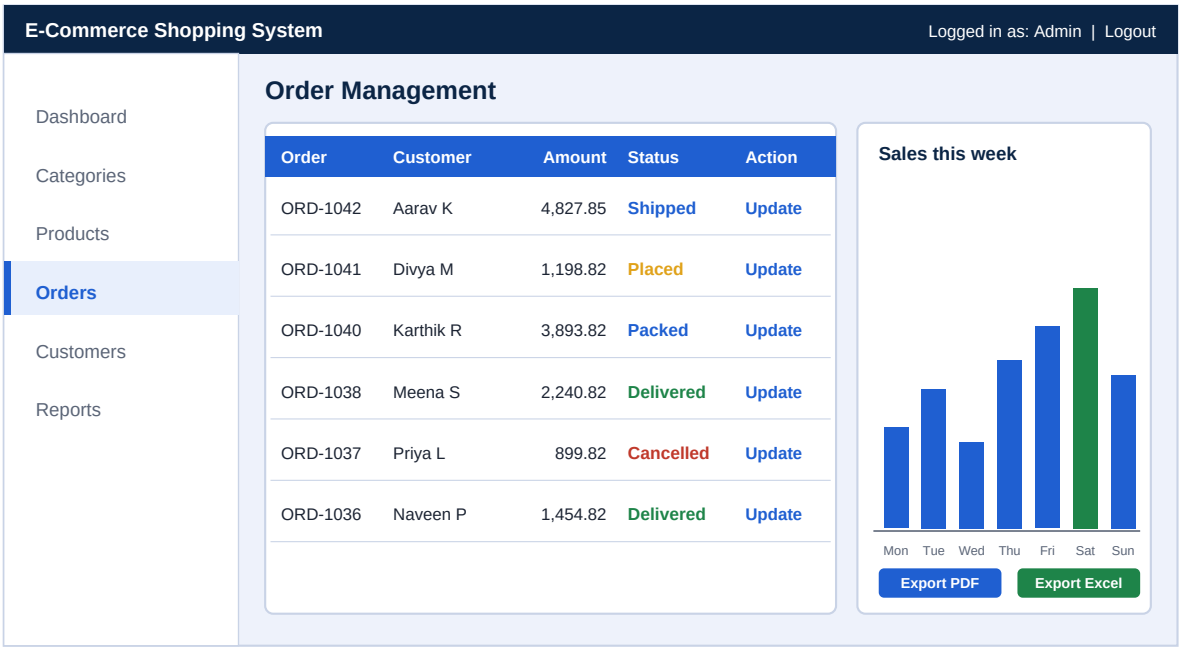



**Figure 9.7:** *Order management and weekly sales report*

## 9.8 Requirement Coverage

The table below maps the functional requirements of Chapter 5 to the screens in which they are realised, showing that every requirement is covered by the application.

Table 9.1: Coverage of functional requirements by application screens

| Requirement | Realised in | Screen |
|---|---|---|
| FR-01 Login and registration | Authentication module | Login page (Figure 9.1) |
| FR-02 Category and product management | Category and product module | Dashboard shortcuts (Figure 9.6) |
| FR-03 Browse, search, filter | Catalogue and search module | Product catalogue (Figure 9.2) |
| FR-04 Shopping cart | Shopping cart module | Cart (Figure 9.3) |
| FR-05 Checkout and address | Checkout and orders module | Checkout (Figure 9.4) |
| FR-06 Stock check and price calculation | Checkout and orders module | Cart and checkout (Figures 9.3, 9.4) |
| FR-07 Payment and confirmation | Payment module | Checkout (Figure 9.4) |
| FR-08 Order history and tracking | Order tracking module | My Orders (Figure 9.5) |
| FR-09 Order and customer management | Order management module | Order management (Figure 9.7) |
| FR-10 Reports | Reports module | Order management and reports (Figure 9.7) |

## 9.9 Results Achieved

Table 9.2: Project objectives and outcomes

| Objective | Outcome |
|---|---|
| Searchable catalogue | Categories and products can be added, updated and found by keyword, category and price |
| Cart and checkout | Customers can fill a cart and complete checkout in a few clicks |
| Accurate pricing | Subtotal, discount, GST, delivery charge and total are computed automatically from the configured rules |
| Reliable stock | Orders that exceed available stock are prevented and stock is reduced transactionally |
| Order tracking | Customers follow status from Placed to Delivered; administrators update status |
| Secure access | Role-based access protects every function |
| Reporting | Sales, stock and order reports are available |

## 9.10 Usability Observations

- **Consistent navigation:** the same header, menu style and colour scheme are used on every page.
- **Colour-coded status:** green, amber, blue and red status labels make order progress recognisable at a glance.
- **Immediate feedback:** validation messages appear as soon as a field or quantity is incorrect.
- **Less manual work:** totals, tax and stock updates are automatic, reducing data entry and errors.
- **Role-aware menus:** customers and administrators see only the functions they are permitted to use.

CHAPTER 10

# 10. Learning Outcomes and Challenges

## 10.1 Technical Skills

- Designing object-oriented solutions using classes, interfaces, inheritance and polymorphism.
- Using collections, generics, file handling and multithreading appropriately.
- Connecting applications to relational databases using JDBC and Hibernate and writing efficient SQL.
- Building web applications with Servlets, JSP and Spring Boot, and exposing REST services.
- Applying authentication and authorisation with Spring Security.
- Writing unit and integration tests with JUnit and debugging with IDE tools.
- Using Git for version control and Apache Tomcat for deployment.

## 10.2 Professional Growth

Beyond technology, the internship built communication, teamwork, time management, problem solving, documentation and professional discipline. Regular discussions with the industry guide, code reviews with colleagues and module-wise planning of the project all contributed to this growth.

## 10.3 Reflection

**Context.** The internship was a valuable month that gave direct exposure to a professional working environment and showed how an organisation expects its members to conduct themselves.

**My role.** As an intern, my responsibility was to follow the organisation’s rules, to learn from the team as a newcomer, and to contribute to the E-Commerce Shopping System within the scope assigned.

**Action taken.** I applied the knowledge and skills from my specialisation to the tasks given, studied new technologies where the project demanded them, and sought guidance whenever a problem remained unresolved.

**Performance.** Colleagues and mentors supported me through every challenge and were available whenever I needed assistance, which helped me progress steadily.

**Gain.** I understood the importance of orientation in settling into a new environment and improved my understanding of individual and group behaviour in an organisation. Technically, I gained confidence in building a complete application end to end.

## 10.4 Challenges and Solutions

Table 10.1: Challenges faced and solutions adopted

| Challenge | Solution adopted |
|---|---|
| Learning many technologies in a short time | Followed the structured plan, practised each concept immediately in small programs and revised from the official documentation. |
| Mapping Java objects to relational tables | Studied Hibernate and JPA annotations and verified mappings against the database design using small test programs. |
| Keeping stock consistent when several customers order the same product | Wrapped order placement in a database transaction that checks and reduces stock together. |
| Correct price, tax and total calculation | Isolated the logic in service methods, used BigDecimal for money and covered the rules with unit tests. |
| Restricting features by user role | Configured Spring Security rules by URL pattern and hid menu options not permitted for a role. |
| Integrating front end and back end | Agreed on clear request and response structures and tested integration module by module. |
| Managing code changes | Used Git with small, frequent commits and discussed changes in code reviews. |

CHAPTER 11

# 11. Conclusion and Future Scope

## 11.1 Conclusion

The internship at Fantasy Solution was a rewarding and well-structured learning experience. In thirty days it covered the journey from the fundamentals of Java to the development of a complete, database-backed web application. Through daily practice, coding exercises and project work, I gained solid skills in object-oriented programming, web development, database connectivity, security, testing and deployment.

The E-Commerce Shopping System developed during the internship demonstrates how these skills combine to solve a practical problem. It replaces manual order taking with a secure and reliable system that manages the catalogue, cart, checkout, orders, stock and reports through a role-based interface. Equally important, the internship taught professional habits: working to a plan, communicating clearly, receiving feedback and documenting work.

I am grateful for the opportunity, and I believe that the experience has prepared me well for the challenges of software development in my future career.

## 11.2 Future Enhancements

- **Payment gateway:** integration with a live payment gateway for secure online payment.
- **Notifications:** automatic e-mail or SMS alerts for order confirmation, shipping and delivery.
- **Wishlist and reviews:** saved products, ratings and customer reviews.
- **Recommendations:** suggestions based on browsing and purchase history.
- **Offers management:** an administrator screen to create and schedule coupons and sales.
- **Multi-vendor support:** several sellers listing products on the same platform.
- **Delivery tracking:** integration with courier services for live shipment status.
- **Mobile application:** a companion Android app for customers and administrators.

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